

Nihal Khatwani

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EDUCATION

Emory University

Expected Grad: May 2027

Applied Mathematics and Statistics (BS), Minor in Computer Science

Atlanta, GA

- **Math Coursework:** Mathematical Statistics I/II, Linear Algebra, Combinatorics, Mathematical Foundations of Data Science, Numerical Analysis, Game Theory
- **Data Science + CS Coursework:** Introduction to Scientific Methods, Introduction to Statistical Computing I/II (R & Python), Regression Analysis, Introduction to Computer Science I/II (Java), Data Structures and Algorithms

EXPERIENCE

Undergraduate Grader

Jan. 2026 – May 2026

Emory University - Department of Data and Decision Sciences

Atlanta, GA

- Graded problem sets and exams for DATASCI 210: Probability and Statistics.
- Provided feedback to students on statistical reasoning, probability concepts, and problem-solving approaches during grading.

Undergraduate Teaching Assistant

Jan. 2025 – May 2025

Emory University - Department of Data and Decision Sciences

Atlanta, GA

- Mentored ~90 students in Python programming, SQL database management and the basics of Git
- Provided hands-on instruction during office hours, covering libraries such as **pandas** and **Matplotlib**.

Project Data Scientist

Aug. 2024 – Dec 2024

The Carter Center

Atlanta, GA

- Links to [Research Poster](#), [GitHub](#)
- Cleaned and analyzed a large-scale social media dataset on conflict to assess US politicians' sentiment.
- Tasked with developing an active learning sentiment analysis model to handle ongoing, high-volume data.
- Manually labeled approximately 150 posts and trained a **logistic regression** model using **scikit-learn**, achieving 78% accuracy in sentiment classification.
- Built a scalable pipeline to integrate iterative learning and improve model accuracy with new data.

Project Analyst

Feb. 2024 – Apr. 2024

Emory Data Science Club

Atlanta, GA

- Link to [Research Poster](#)
- Developed a Python script to make curve-grading more efficient for undergraduate Biology 141 and 142 labs.
- Utilized **pandas** and **NumPy** to discern outliers and filtered respective scores to normalize data.
- Adjusted data by assigning *z*-scores and percentages to individual test scores to account for statistical outliers and anomalies. Fit data with normal and logit-normal distributions to get parameter estimates.

PROJECTS

Live Traffic & Wait Times with a Temporal Fusion Transformer

Aug. 2025 - Sep. 2025

- Used OpenStreetMap road data to train a Temporal Fusion Transformer (TFT) that predicts bus delays 10 min ahead (test error ≈ 0.78 min MAE).
- Added percentile estimates so the map can show best/likely/worst cases; backtested to ensure output.
- Shipped a lightweight FastAPI service and Leaflet web map showing per-stop risk using a control simulation.

Housing Price Classification with Random Forest

Oct. 2024 - Dec. 2024

- Developed a Random Forest Classifier to predict housing prices as "above" or "below" median, achieving an accuracy of 77% after revisiting feature selection and re-balancing classes.
- Performed data preprocessing and feature engineering, transforming continuous price data into binary classes.

SKILLS

Languages: Python, SQL, R, Java, MATLAB

Developer Tools: Jupyter Notebook, JupyterLab, VS Code, PyCharm, MySQL, Tableau, Power BI

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, Keras, PyTorch, NLTK, spaCy, SciPy

Concepts: LDA, PCA, NLP, Dashboarding, Linear Algebra, Calculus, Probability and Statistics, Model Evaluation, Parameter Tuning, Data Cleaning, Data Visualization